

Stock and Option Modelling

A project report
submitted by

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0.1 Introduction - Company Info

\$AMD

Advanced Micro Devices Inc. manufactures processors, graphics cards, and embedded systems. Over the last 5 years, AMD has been extremely volatile. With the lowest price being around 10\$ a share in early 2018, while the all-time-high was around \$161 which occurred in Nov. 2021. AMD has gained a foothold in the market due to its high performing products. In Q1 2023, they released the AMD Ryzen 9 7950X3D, this CPU is considered one of the fastest gaming CPU's on the market, competing with Intel's newest i11 series chip set. While movement in stock price can occur for a multitude of reasons, the trends are a mix of up and down, with potential for a lot of upside, and inversely, severe draw down. The following five year chart gives an idea for how volatile AMD can be, notice the low occurring in 2018 and the all time high occurring during the fall of 2021:



\$TNET

TriNet is a professional employer organization that provides small and medium size businesses with full service HR solutions tailored by industry. They mainly provide payroll services, but also support services like business benefits and risk mitigation advice. In 2022, they brought in a total revenue of \$4,885,000. They are a profitable, public company which brought in \$647,000 EBITDA (expenses before interest, taxes, depreciation and amortization– attempts to represent cash profit generated by the company’s operations.). They performed well in 2022, which correlates to the increase in demand for workers across all industries, with the main exception being the tech industry. \$TNET is considered a relatively stable stock that mostly trades sideways with a slight uptrend. The exception to this is the pandemic era. Observe the following five year chart which displays this:



\$SNAP

Snap Inc. is a company that hosts the social media platform, SnapChat. Their stock history is particularly interesting over the course of the pandemic. The low coming at the beginning of the pandemic at around \$8 a share, but less than a year later the stock shot up to \$83, with very little draw down along the way. The general sentiment at that point was they were overvalued, and investors started to lose faith as their user growth started to slow. The year following the high of \$83, they dropped down to a low of about \$7.80. During Q1 2023, there has not been any major news relating to snap. This is reflected by its tight band trading sideways between \$8-\$10.



0.2 Stock Performance

Ticker	Avg Daily Absolute Change	Avg Positive Returns	Avg Negative Returns	% of positive days	Volatility**
\$AMD	\$2.65	+\$2.66	-\$2.67	46.43%	14.26%
\$TNET	\$1.26	+\$1.26	-\$1.26	47.40%	5.2%
\$SNAP	\$0.81	+\$0.75	-\$0.87	46.42%	51.94%

As we can see from this data, all three stocks have a higher average negative return compared to the lower average positive return counterpart. In addition to that, all stocks have more losing days than positive days. Those are pretty key metrics as it seems that all stocks did not perform well in Q1 2023.

** daily volatility, not annualized

0.3 Black-Scholes Derivations

Recall the black-scholes formula given by:

$$c = S(0) \phi(w) - Ke^{(-rt)} \phi(w - \theta\sqrt{T})$$

where

$$w = \frac{\ln(S/K) + (r + \theta^2/2) T}{\theta\sqrt{T}}$$

and

K is the strike price.

S(0) is the initial stock price.

r is the risk free rate.

T is the time until expiration.

θ^2 is the annualized volatility.

\$AMD

The following is the black-scholes derivation specific for \$AMD. The call option was bought on March 27th 2023, and has an expiry of April 21st. For this option, the parameters are: $S(0) = \$96.61$, $K = \$100$, $T = 19/252$, $\theta^2 = 0.2707696$, $r = 0.0353$

$$w = \frac{\ln(96.61/100) + (0.0353 + 0.2707696^2/2) 19/252}{\sqrt{19/252 * 0.2707696}} = -0.151306$$

$$c = 96.61\phi(-0.151306) - 100e^{(-0.353*19/252)}\phi\left(-0.151306 - \sqrt{19/252 * 0.2707696}\right) = \mathbf{\$4.17}$$

\$TNET

The following is the black-scholes derivation specific for \$TNET. The call option was bought on March 27th 2023, and has an expiry of April 21st. For this option, the parameters are: $S(0) = \$76.74, K = \$85, T = 19/252, \theta^2 = 0.1310247, r = 0.0353$

$$w = \frac{\ln(76.74/85) + (0.0353 + 0.1310247^2/2) 19/252}{\sqrt{19/252 * 0.1310247}} = -0.952057$$

$$c = 96.61\phi(-0.952057) - 85e^{(-0.353*12/252)}\phi\left(-0.952057 - \sqrt{19/252 * 0.1310247}\right) = \mathbf{\$0.66}$$

\$SNAP

The following is the black-scholes derivation specific for \$SNAP. The call option was bought on March 27th 2023, and has an expiry of April 21st. For this option, the parameters are: $S(0) = \$11.74, K = \$14, T = 19/252, \theta^2 = 0.4340935, r = 0.0353$

$$w = \frac{\ln(11.74/14) + (0.0353 + 0.4340935^2/2) 19/252}{\sqrt{19/252 * 0.4340935}} = 0.176506$$

$$c = 96.61\phi(0.176506) - 14e^{(-0.353*12/252)}\phi\left(0.176506 - \sqrt{19/252 * 0.4340935}\right) = \mathbf{\$0.20}$$

The following table represent the difference between the black-scholes calculations and actual prices.

Ticker	Black-Scholes Price	Actual Price	% difference
\$AMD	\$4.17	\$3.4	22.6%
\$TNET	\$0.66	\$1.65	60%
\$SNAP	\$0.20	\$0.46	56.52%

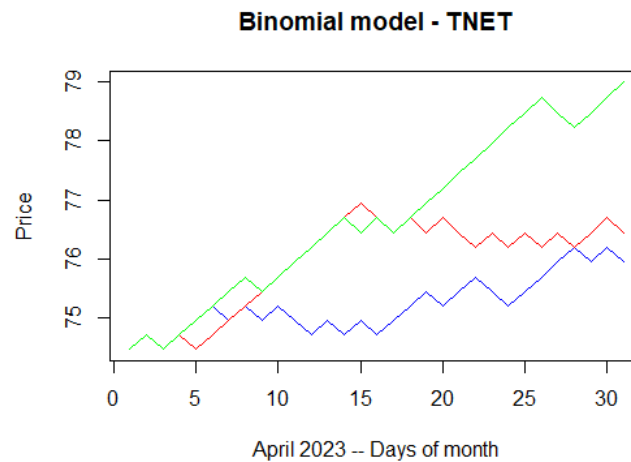
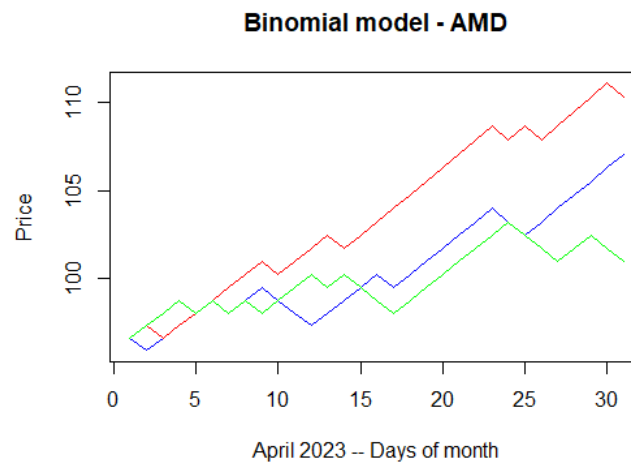
One can see that the % difference between the two prices are quite high. To produce more accurate results, rather than using the previously calculated volatility, perhaps we can use the implied volatility listed in the data set. Rerunning the same derivations with modified θ^2 yields the following results.

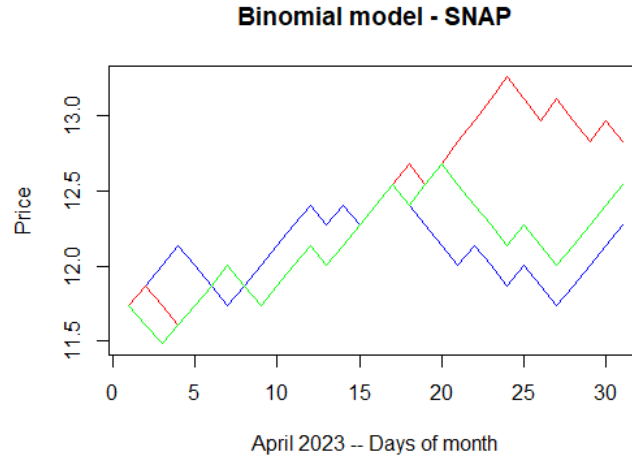
Ticker	θ^2	Black-Scholes Price	Actual Price	% difference
\$AMD	0.22877089	\$3.72	\$3.4	9.4%
\$TNET	0.29398084	\$1.81	\$1.65	9.7%
\$SNAP	0.87160896	\$0.48	\$0.46	4.35%

With a modified θ^2 , we can see our results have increased in accuracy by orders of magnitude. This indicates the output of the black-scholes equation is heavily dictated by the volatility. This correlates with the fact that θ^2 is used three times in the black-scholes equation.

0.4 Simulations: Binomial Model

All three stocks were ran against binomial simulations to predict the future value of their respective call. Below is a visual representation of each stock for how they could potentially perform. Each color of the line indicates a different simulation that was ran. For computing purposes, only simulations with three iterations were graphed.





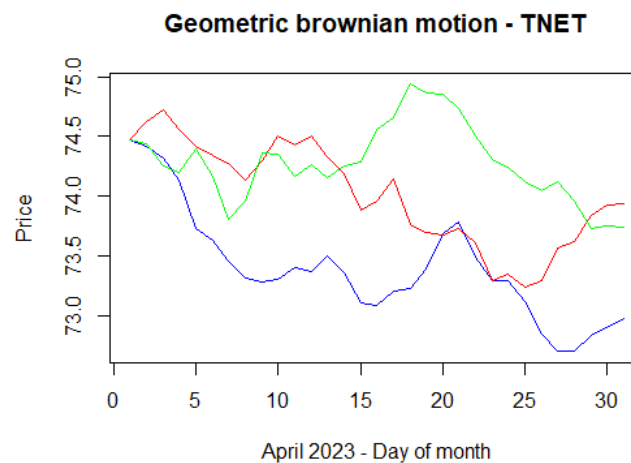
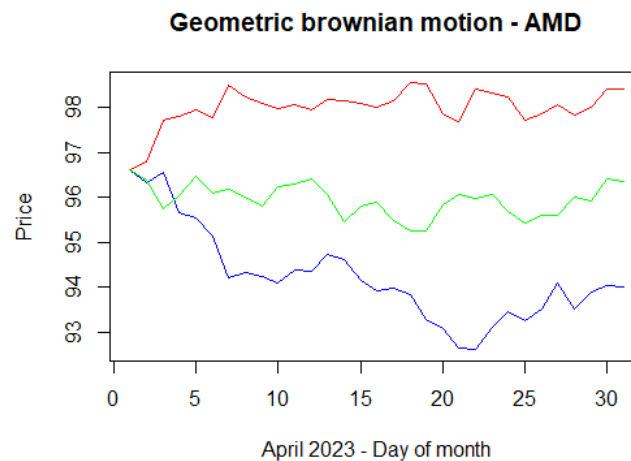
Furthermore, the table below represents a price analysis of the stocks that were ran against the binomial simulations.

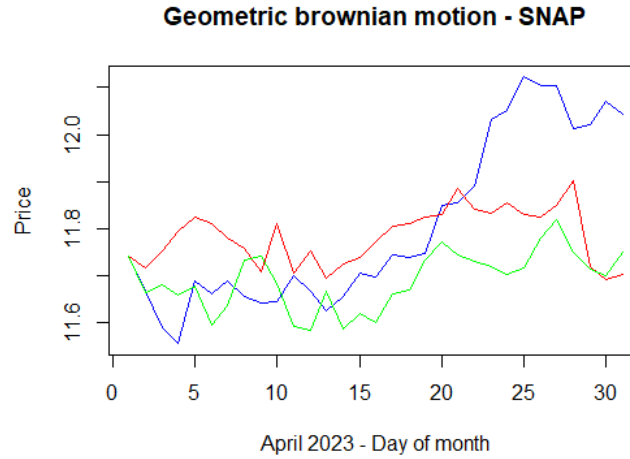
Ticker	Predicted Price	Average Distance From Strike	Iterations
\$AMD	\$78.36	-\$6.64	1,000
\$AMD	\$78.37	-\$6.62	100,000
\$TNET	\$12.45	-\$1.55	1,000
\$TNET	\$12.45	-\$1.55	100,000
\$SNAP	\$96.69	-\$3.3	1,000
\$SNAP	\$96.69	-\$3.3	100,000

While all simulations for all stocks produced an average loss, it is noteworthy to indicate that a high amount of iterations does not seem to have an affect on the predicted price. Furthermore, running binomial simulations against various stocks is a great exercise to get a handle on how these simulations behave, however, it is made apparent that these models are not a great predictor for stock price. The contrast of price movement is shown between the binomial simulation graphs above and the actual price graphs referenced in section I of this report.

0.5 Simulations: Continuous Model Using Geometric Brownian Motion

All three stocks were ran against a geometric brownian motion model to predict the future value of their respective call. Below is a visual representation of each stock for how they could potentially perform. Each color of the line indicates a different simulation that was ran. For computing purposes, only simulations with three iterations were graphed.





Furthermore, the table below represents a price analysis of the stocks that were ran against the continuous simulations.

Ticker	Predicted Price	Average Distance From Strike	Iterations
\$AMD	\$76.72	-\$8.28	1,000
\$AMD	\$78.07	-\$6.92	100,000
\$TNET	\$11.30	-\$2.69	1,000
\$TNET	\$11.74	-\$2.26	100,000
\$SNAP	\$94.65	-\$5.35	1,000
\$SNAP	\$96.33	-\$3.67	100,000

While none of the simulations provided an average profit, the graph of the simulated price looks to be much more representative of how an actual stock price behaves. Additionally, in contrast to the binomial model, the iteration count of the continuous model does produce noticeably different results. This model seems to provide a good starting basis for predicting how a given stock can behave over a period of time. However, to actually predict and trade profitably, much more strategic thought and work is required to create a model that is considered to be profitable.

0.6 Comparisons & Final Thoughts

The table below represents the difference between the average call value of the geometric brownian motion model against the close price when the call expired on April 21st, 2023. The average price of a call was calculated by determining the value of the call at the end of an individual simulation, and then computing the average of all those values. While most of the simulations produced a value of \$0, there were some simulations that provided a profit.

Ticker	Average Call Value	Actual Close Price	Strike Price	Actual Call Value
\$AMD	\$5.44	\$88.43	\$100	\$0.0
\$TNET	\$0.08	\$85.54	\$85	\$0.54
\$SNAP	\$0.005	\$10.01	\$14	\$0.0

The results from this table can be misleading, when analyzing the average call value against the actual call value, \$SNAP and \$TNET ended up being pretty close, while we are very incorrect with \$AMD. We should keep in mind two main thoughts: the average call values are all pretty low relative to the corresponding stock price, this can indicate a high risk if actually trading off these values. In addition to that, average call value does not account for average loss. Refer to the tables in sections V and VI to get an idea for what the predicted price was for each stock. Therefore, given the results of the simulations and subsequent predictions, it would seem beneficial to buy a put as opposed to buying a call. Either way, the results concluded from this project highlight how difficult it can be to predict a given share price and subsequent call value.